

Yang Shaopeng

Personal Information:

- Basic Info: Male | Date of Birth: Mar 1989 | Residence: Wuxi, Jiangsu
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 - Education:
 - Master's Degree (Electronic and Communication Engineering), Jiangnan University, 2014
 - Bachelor's Degree (Electronic Information Engineering), Jiangnan University, 2011
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Professional Summary:

- 10+ Years Cross-Domain Experience: Encompassing Industrial IoT (IIoT), Big Data, and Machine Learning/LLM applications, proficient in the full Cloud-Pipe-Device stack.
 - Technology Integration Skills: Combines software/hardware development (Python/Java/.NET, Edge Linux), data analysis (SQL/R), and project management (Siemens Certified Project Manager).
 - Industry Practice for Digitalization: Deeply involved in digital solutions for semiconductor & optical communication manufacturing/test development, automotive manufacturing & software (Telematics, Predictive Maintenance, Catena-X Data Space), healthcare, energy, etc. Holds multiple patents and has experience delivering multinational projects.
 - Language: Fluent in English (Cambridge BEC Higher, TOEIC 980), with experience in multinational collaboration and technical presentations.
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Work & Project Experience:

► 2021.06 – Present | Siemens Ltd., China, Wuxi Innovation Center | Expert Engineer

- Data Collection, Modeling & Analytics Architect
 - Industrial IoT Predictive Maintenance Solution (Python, Vue.js, SQL, Jupyter, Nginx, AI)
 - Responsible for architectural design and core coding implementation.
 - Collected big data via sensors (temp, humidity, vibration, audio), performed algorithmic modeling for feature extraction, delivering predictive monitoring solutions.
 - Involved IoT big data modeling/analysis, algorithm processing, feature extraction for industrial IoT predictive maintenance scenarios.
 - Solution deployed in automotive manufacturing, healthcare, drilling, and other manufacturing industries.
 - Integrated LLMs to develop applications like Intelligent Data Assistants and Knowledge Bases.
 - Urban IoT Weather Big Data & Video Surveillance Project (Python, SQL, RabbitMQ, Kafka)
 - Responsible for architectural design and core implementation.
 - Collected massive data via sensor networks covering Wuxi's Xinwu District, processed in the cloud for real-time monitoring.
 - Camera applications used for Smart Parking. A government collaboration project successfully deployed on the Feifeng Platform in Xinwu District.
 - Urban IoT Bridge Big Data Monitoring Project (Python, SQL, Jupyter, Nginx)
 - Responsible for project architecture design, development, and deployment.
 - Collected extensive bridge vibration data via installed vibration sensors and cameras, modeled data to extract vehicle features, enabling real-time monitoring of large/heavy trucks.
 - Project successfully passed acceptance for the Wuxi Sci-Tech Innovation Project.
 - Large Language Model (LLM) Pre-research & Application Development
 - Responsible for architecture design and core code implementation.
 - Researched LLM use cases and implementation strategies integrated with Siemens IoT Cloud Platform, covering competitor SWOT analysis, framework comparison, and implementation paths.

- Supported business units in successfully organizing Hackathons and customer case applications; delivered multiple internal technical sharings.
 - Applied LLMs to intelligent data analysis and knowledge base assistants.
- First Successful Domestic Deployment of Catena-X Automotive Data Space
 - Catena-X is a German-led European automotive industry data exchange standard.
 - Tracked and participated in China region meetings since 2021, researching the technical framework and attempting deployment.
 - Successfully deployed a Data Space exchange PoC between CATL (Ningde) headquarters and BMW Germany within a short timeframe (3 days) in late 2024, covering data exchange, battery passport, and digital twin, marking the first successful domestic implementation.
- Project Manager
 - Xinwu District Sci-Tech Project Delivery
 - Managed the annual Xinwu District Sci-Tech project lifecycle: project selection, delivery, results acceptance, and fund application, achieving successful project acceptance.
 - Coordinated participation in the Wuxi Annual IoT Expo.
 - Decoupling Cross-Departmental Communication
 - Led the "Decoupling" project, focusing on assessing the relevance of internal technologies within the context of Sino-US tech decoupling.
 - Managed cross-departmental communication, collaborating with technical experts and department heads to identify common technologies and potential risks, analyzing their correlation with decoupling trends.
 - Project covered multiple technical domains, requiring strong comprehensive technical understanding and efficient communication skills.
 - Final deliverable was a comprehensive technical risk assessment report with targeted improvement suggestions, providing key input for organizational technology strategy.
 - AirLink Porting Feasibility Project Delivery
 - Led a team (6 members) to complete the feasibility study and attempt to port Siemens AirLink from Windows to Linux, particularly utilizing containerization technologies like Docker.
 - Successfully delivered a validated Demo and feasibility report, proposing further optimization solutions using Docker container networking and Privilege mode.
- Communication contact
 - Acted as primary contact for projects involving colleagues in Germany, other European countries, and India. Independently managed the interface for software/hardware upgrades of key equipment at a specific Siemens factory in China.
 - Responsible for planning and organizing the annual "Open Source@西门子" event (preparing for the 3rd iteration). Promoted Siemens' progress and activities in industrial open source, inviting guests from other fields (Tsinghua university, Xenomai real-time control, AI, open-source semiconductor, RISC-V, etc.). Presented as a speaker on Siemens LLM API solutions (opensource.siemens.com) last year and speaker for Catena-X OSS implementation this year.
- **2020.06 – 2021.06 | Bosch China Innovation and Software Development Center | Competence Leader**
- UAES Tool Test Development & Telematics Big Data Project Manager (Jira, Java, Python, Kafka, RabbitMQ, Vue.js)
 - Liaised with the Telematics big data development team to build data collection, analysis, and visualization services.
 - Managed tool and test development, CI/CD requirements from various UAES business units, ensuring project delivery.
- Wuxi Bosch Telematics Big Data Project Interface & Architect(Python, Vue.js, MQTT, Java, OSGi)

- Served as project interface for cloud-side Telematics big data development needs, including architecture design and coding.
- Built the MQTT protocol architecture and PKI management system.
- Participated in device-side middleware software design and coding based on the OSGi framework.

➤ **2018.05 – 2020.05 | Bosch Rexroth (Changzhou) Co., Ltd. | Sr. Software Engineer**

- UWB-based Indoor Real-Time Location & Bosch Production System Software Product R&D & Test System Setup (.NET, Python)
 - Core team member responsible for real-time location algorithm, big data processing and modeling, integrating it into the Bosch Production System.
- Sino-German Government Cooperation Project: 3D Real-Time Location-based Logistics Optimization & Collision Avoidance (.NET, Eclipse Mita, C)
 - Core developer responsible for software product R&D for real-time machine status data collection based on IoT kits.

➤ **2014.07 – 2018.04 | Finisar (Wuxi) Photonics Technology Co., Ltd. | Sr. Engineer**

- Production Big Data-based Quality Monitoring System Development (.NET, R, FineUI, SQL)
 - Integrated production big data with 6Sigma theory, using .NET and R to develop quality monitoring tools/platforms like GRR, SPC, and real-time dashboards, achieving production transparency based on big data.
- Optical Module Test System Software Development & Software Test Platform Localization Maintenance (.NET, LabWindows CVI)
 - Developed customized optical module test procedure software for test big data collection and algorithm validation.
 - Served as liaison with the US branch for production data transfer and test maintenance, and with the Malaysia development team for localization maintenance and configuration.